

Create a Web Database in App Inventor 2

On our App Inventor journey, which we embarked upon many months ago, we now move on to creating a Web database using App Inventor 2. This tutorial takes you step by step into the practical details and the nitty-gritty of creating the Web database.



I hope all of you are learning a few things through these Android tutorials, and I am so glad to hear from some of you. Continuing the series, we have another interesting topic to cover—an online database.

So far we have learned how to keep or store things locally on your mobile phone or, say, TinyDB. But have you thought of storing the data virtually to some remote location and making it available to all the connected devices? All users should have the permission to access this database or the database security should be public. In this tutorial, we will set up an online database and make it accessible across devices.

Before we move ahead, let me first introduce you to the database. In our application, we are going to use the Firebase database, which is a Google provided cloud database, and the good thing is that it is open source and freely available for use.

Features of Firebase

1. Firebase uses the GUI based data centre, so that you can see all the entries and manage them as well.
2. Being on the cloud, it is accessible on the go to any of the devices.
3. Smart authentication provides secure access.
4. The real-time database reflects changes as soon as the data is updated.
5. You can add collaborators so that many people can manage the database.
6. No coding is needed for fetching and storing data.

Creating a project in the Firebase database

1. To use Firebase, you must have a working Google account, which will also be needed to run App Inventor.
 2. Type <https://console.firebase.google.com/> in the address bar of the browser you are using and hit *Enter*.
 3. If asked for login credentials, give your Google account credentials and proceed.
 4. Follow the on-screen instructions, which include accepting the terms, etc.
 5. Once everything is done, you will see the screen shown in Figure 1.
 6. Click on the *Create New Project* button as shown in Figure 1.
 7. Give the name of the project and country/region as shown in Figure 2.
 8. Once your project is created, you will see the dashboard of your application. It will be similar to what is shown in Figure 3.
 9. Now select *Database* from the left hand palette and then *Rules* from the next page. It will display something like what is shown in Figure 4.
 10. This time, we won't give access to the database to all users; hence, we need to change the rules. You can manually change the rules by typing values, as shown in Figure 5. Once this is done, click on the *Publish* button.
- You have, now, successfully created an empty space for your project in the Firebase database, and have given read and